Jessie Yuan

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Education

Carnegie Mellon University (CMU) – *Pittsburgh, PA* Aug. 2022 – May 2026
B.S. in Computer Science, Minor in Mathematics

- GPA: 4.0/4.0
- **Relevant Coursework:** Robot Learning (16-831)*, Fundamentals of Human-Robot Interaction (16-701)*, Machine Learning (10-701), Embodied AI Safety (16-886), Algorithm Design and Analysis (15-451)

Research Experience

Interactive and Trustworthy Robotics (Intent) Lab – *Robotics Institute, CMU* May 2025 – Present
Undergraduate Researcher, *Advised by Prof. Andrea Bajcsy*

- Led research on identifying and reacting to uncertainty in vision-language models (VLMs) during policy steering for robotic manipulation tasks.
- Implemented system using world models and VLMs to translate rollouts from a diffusion policy into textual narrations and select appropriate robot behaviors based on nuanced language instructions.
- Applied conformal prediction to quantify VLM uncertainty, distinguishing between scenarios where multiple valid behaviors exist versus when the base policy cannot accomplish the task.
- Tested various methods to improve confidence score quality, ultimately implementing marginalization over user intent to explicitly model ambiguity in human commands.

Robotic Caregiving and Human Interaction (RCHI) Lab – *Robotics Institute, CMU* Feb. 2023 – May 2025
Undergraduate Researcher, *Advised by Prof. Zackory Erickson*

- Conducted in-depth literature reviews on LLM integration with robotics and wearable sensors for detecting eating behaviors to guide project frameworks and methodologies.
- Developed a ChatGPT-driven speech interface for a commercial feeding robot through prompt engineering and awarded CMU Summer Undergraduate Research Fellowship for this project.
- Designed, organized, and conducted three human studies, including procedure documentation, participant interaction, and data analysis to collect training data and evaluate system efficacy and user experience.
- Designed a ROS-based system to collect, label, and process data from wearable sensors; cleaned, visualized, and performed feature extraction on this data; and trained and deployed binary classification and regression models using PyTorch.

Publications

A. Padmanabha*, **J. Yuan***, T. Mehta, R. K. Jenamani, E. Hu, V. de León, A. Wertz, J. Gupta, B. Dodson, Y. Yan, C. Majidi, T. Bhattacharjee†, and Z. Erickson†, "WAFFLE: A Wearable Approach to Bite Timing Estimation in Robot-Assisted Feeding", accepted to *The ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, 2026.

A. Padmanabha*, **J. Yuan***, J. Gupta, Z. Karachiwalla, C. Majidi, H. Admoni, and Z. Erickson, "VoicePilot: Harnessing LLMs as Speech Interfaces for Physically Assistive Robots", in *The ACM Symposium on User Interface Software and Technology (UIST)*, 2024. **Oral Presentation.**

J. Yuan, J. Gupta, A. Padmanabha, Z. Karachiwalla, C. Majidi, H. Admoni, and Z. Erickson, "Towards an LLM-Based Speech Interface for Robot-Assisted Feeding", in *Adjunct Proceedings of the ACM Symposium on User Interface Software and Technology (UIST Adjunct)*, 2024. **Live Demonstration.**

Work Experience

Scilligence – *Cambridge, MA* Jun. 2021 – Aug. 2022
Software Development Intern

- Contributed to both the frontend and backend development of a cutting-edge web-based lab notebook, utilizing HTML, CSS, JavaScript, C#, and SQL.

- Led a project to develop LCD digit recognition software using OpenCV in Python and presented the project to the entire development team, effectively communicating its goals and applications.

Teaching Experience

Teaching Assistant for 21-128: Mathematical Concepts and Proofs, *CMU*	Fall 2024
Teaching Assistant for 15-251: Theoretical Computer Science, *CMU*	Spring 2024, Fall 2023
Teaching Assistant for 15-122: Imperative Computation, *CMU*	Spring 2023